Conclusion: BLOCKED

The task **“Prove genuine work and storage claims pass before reusing reward templates”** is **BLOCKED**.
A fresh verification of current environment and workspace access confirms that none of the seven required cryptographic and governance gates are satisfied in the accessible project files. No code, verifier, reward logic, economic sentinel, or environment configuration has been modified.

Current Access Verification & Search Scope

Before evaluating the seven evidence gates, current access was verified across all local and authorized cloud sources:

1. **Local Execution Sandbox (/home/bard, /tmp, and root filesystem)**: **OBSERVED** — Accessible, but contains zero project repository files. Direct path checks for docs/mining/report-template-proof-validation.md, scripts/report-template-proof-lab/policy.test.mjs, docs/mining/candidate-proof-system-design.md, docs/mining/proof-artifact-cloud-search.md, and docs/mining/proof-and-sustainable-economics-research-brief.md all returned False, and a full filesystem search for report-template, candidate-proof, policy.test, sustainable-economics, and proof-artifact returned no matches.
2. **Authorized Google Drive Workspace Corpus**: **OBSERVED** — Accessible for three indexed project archives owned by PJJRA Strongman (jjrasltr@gmail.com):
    - ReplitExport.zip (id=17Ek59rwfYJbtFV_BM3GktjhCtJrKXLQT, updated 2026-08-24T16:50:27-04:00)
    - ReplitExport-2.zip (id=1jD8iXvx6BXE9ydsOiyC9KbgaOQAf-QfM, updated 2026-10-08T03:44:37-04:00)
    - rainbowjobs94-ltbeta-e6afdc26-8a5edab282632443_1772017269522.txt (id=1jyDuEYnzMFJw6L7yvCjmUYMNAfAfq7ec, updated 2026-03-25T19:35:32-04:00)
3. **Privacy & External Transmission Compliance**: No private documents or workspace contents were sent to any external AI or third-party service.

Evidence Matrix

Gate / Artifact Requirement	Status Label	Verified Findings & Cryptographic Evidence	Source Citation / Revision / Artifact Hash
Target File: docs/mining/report-template-proof-validation.md	**UNKNOWN**	File is not present in the local runtime sandbox or in the three indexed Google Drive project exports (ReplitExport.zip, ReplitExport-2.zip,	Sandbox check (False); Drive exports

Gate / Artifact Requirement	Status Label	Verified Findings & Cryptographic Evidence	Source Citation / Revision / Artifact Hash
		rainbowjobs94-ltbeta-e6afdc26-8a5edab282632443_1772017269522.txt).	
Target File: scripts/report-template-proof-lab/policy.test.mjs	**UNKNOWN**	File is not present in the local runtime sandbox or in the three indexed Google Drive project exports. In package.json, the test runner is configured only for vitest run over src/**/*.{test,spec}.{ts,tsx}.	Sandbox check (False); package.json & vitest.config.ts in rainbowjobs94-ltbeta-e6afdc26-8a5edab282632443_1772017269522.txt
Target File: docs/mining/candidate-proof-system-design.md	**UNKNOWN**	File is not present in the local runtime sandbox or in the three indexed Google Drive project exports.	Sandbox check (False); Drive exports
Target File: docs/mining/proof-artifact-cloud-search.md	**UNKNOWN**	File is not present in the local runtime sandbox or in the three indexed Google Drive project exports.	Sandbox check (False); Drive exports
Target File: docs/mining/proof-and-sustainable-economics-research-brief.md	**UNKNOWN**	File is not present in the local runtime sandbox or in the three indexed Google Drive project exports.	Sandbox check (False); Drive exports
1. Exact Verifier & Circuit (version, revision, file hashes)	**OBSERVED** (Absence in codebase) / **UNKNOWN** (External)	**OBSERVED**: package.json across ReplitExport.zip and rainbowjobs94-ltbeta-e6afdc26...txt contains UI/web dependencies (react, @supabase/supabase-js, zod, lucide-react) and zero cryptographic proving/verification packages (snarkjs, circomlib, ffjavascript, or WASM/R1CS verifiers). **UNKNOWN**:	package.json in ReplitExport.zip (id=17Ek59rwfYJbtFV_BM3GktjhCtJrKXLQT) and rainbowjobs94-ltbeta-e6afdc26-8a5edab282632443_1772017269522.txt

Gate / Artifact Requirement	Status Label	Verified Findings & Cryptographic Evidence	Source Citation / Revision / Artifact Hash
		Exact .circom, .r1cs, .wasm, or verifier contract revision and SHA-256 hashes.	
2. Public Verification Key (provenance & hash)	**UNKNOWN**	No verification_key.json, .zkey, Powers of Tau ceremony transcript, beacon hash, or verification key digest exists in the accessible workspace archives.	ReplitExport.zip; ReplitExport-2.zip; rainbowjobs94-ltbeta-e6afdc26...txt
3. Precise Compute & Storage Statements (units, inputs, roots/chunks, challenge rules)	**OBSERVED** (Simulated UI only) / **UNKNOWN** (Cryptographic statement)	**OBSERVED**: src/components/platform/FileMiningTab.tsx contains no proof statement; it computes a front-end formula from user-input fileSizeMB (baseCost = size * 0.50, currentBlockSpeed = 1247, shiftCoinRate = 0.0847, nearestNodeFactor = 1.15) with the comment // Simulated blockchain data - in production this would come from actual protocol. src/hooks/usePostMining.ts uploads files to Supabase Storage bucket postmining (500 MB limit) and inserts metadata into public.post_mining_content without chunking, Merkle root commitment, or challenge-response verification. **UNKNOWN**: Approved mathematical constraint	src/components/platform/FileMiningTab.tsx and src/hooks/usePostMining.ts in ReplitExport.zip and ReplitExport-2.zip

Gate / Artifact Requirement	Status Label	Verified Findings & Cryptographic Evidence	Source Citation / Revision / Artifact Hash
		equations for compute work or storage possession.	
4. Public-Input Schema & Byte-Level Encoding (field order, widths, endianness, normalization)	**UNKNOWN**	No public-input signal array specification, prime field modulus normalization rule (e.g., BN254 scalar field r), byte-width table, or little/big-endian serialization schema is present in the accessible archives.	ReplitExport.zip; ReplitExport-2.zip; rainbowjobs94-ltbeta-e6afdc26...txt
5. Independently Sourced Positive Vectors (compute & storage inputs, proofs, expected results, provenance)	**UNKNOWN**	No genuine Groth16/plonk proof objects (pi_a, pi_b, pi_c), public signal arrays, raw storage chunks, Merkle sibling paths, or provenance records exist in the accessible files. (Per instructions, no synthetic or mock vectors have been created or substituted.)	ReplitExport.zip; ReplitExport-2.zip; rainbowjobs94-ltbeta-e6afdc26...txt
6. Negative Tests (tampered inputs/chunks/indices/roots/paths/proof bytes & replay/alternate encoding rejection)	**UNKNOWN**	No cryptographic negative test suite or nullifier/canonical-encoding replay guard is present in the accessible archives. The only SQL security migration present (20260822161903_b478971b-253c-4c72-bfc2-7fcf9329e355.sql) restricts PostgreSQL Row-Level Security (RLS) policies on public.post_mining_content, public.post_reactions, and public.connections.	supabase/migrations/20260822161903_b478971b-253c-4c72-bfc2-7fcf9329e355.sql in ReplitExport.zip

Gate / Artifact Requirement	Status Label	Verified Findings & Cryptographic Evidence	Source Citation / Revision / Artifact Hash
7. Explicit Owner Approval of Exact Artifact Versions, Hashes, Statements & Encodings	**UNKNOWN**	No signed or explicit owner approval record locking specific circuit versions, SHA-256 artifact hashes, constraint statements, and canonical byte encodings was found in the accessible workspace.	ReplitExport.zip; ReplitExport-2.zip; rainbowjobs94-ltbeta-e6afdc26...txt

Exact Owner Decisions and Artifacts Needed to Unblock

To transition this task from **BLOCKED** to **READY**, please supply or authorize access to the following exact items:

1. **Workspace Access to the Mining Proof Lab Files**:
   - Provide or grant Google Drive / repository access to the archive or branch containing docs/mining/report-template-proof-validation.md, scripts/report-template-proof-lab/policy.test.mjs, docs/mining/candidate-proof-system-design.md, docs/mining/proof-artifact-cloud-search.md, and docs/mining/proof-and-sustainable-economics-research-brief.md.
2. **Frozen Circuit & Verifier Package with Cryptographic Digests**:
   - Supply the exact compiled circuit and verifier artifacts (e.g., .circom source, .r1cs, prover .wasm, and verifier module/contract) along with their exact version identifiers, source commit revision, and SHA-256 file hashes.
3. **Public Verification Key & Ceremony Provenance**:
   - Supply the exact verification_key.json (and corresponding .zkey provenance, Phase-1 Powers of Tau transcript, and Phase-2 circuit-specific contribution hashes) together with its SHA-256 digest.
4. **Formal Compute & Storage Statement Specification**:
   - Approve the exact mathematical relation proven by the circuit:
     - **Compute statement**: Exact hash/work function, iteration/step count, difficulty threshold or target, input commitment, and output digest.
     - **Storage statement**: Exact chunk size in bytes, tree arity, hash function (e.g., Poseidon vs. SHA-256), Merkle root construction, leaf index derivation from the challenge seed/beacon, and epoch/window validity rules.
5. **Canonical Public-Input Schema & Anti-Malleability Encoding Rules**:
   - Specify the exact ordered array of public signals (publicSignals[0..n]), bit/byte widths, endianness, field-element reduction rules modulo the curve scalar field r, and canonical nullifier construction so that alternate encodings (e.g., unreduced field elements x + r, padded hex strings, or reused nonces) are deterministically rejected before reward template reuse.
6. **Independently Sourced Positive & Negative Proof Vectors**:
   - Provide at least one genuine, independently generated positive compute proof

vector and one genuine positive storage proof vector (with full witness inputs, public signals, proof object, and generator revision), plus negative test vectors covering mutated public inputs, wrong Merkle sibling paths/indices, corrupted proof bytes, and replayed/alias-encoded work claims.

7. **Explicit Owner Sign-Off**:
    - Provide explicit written approval binding the exact SHA-256 hashes of the circuit, verification key, public-input schema, and test vectors listed above.

(Note: Because the approval gates above are not yet satisfied, no implementation or code/test modification plan is proposed or executed at this stage.)

Cautious Fair-and-Sustainable Economics Summary

Based strictly on the verified source code in ReplitExport.zip, ReplitExport-2.zip, and rainbowjobs94-ltbeta-e6afdc26-8a5edab282632443_1772017269522.txt, the platform's current economic implementation is structured as follows:

1. Verified Evidence: Collected Customer Revenue vs. Internal Balances & Simulated Values

- **Collected Customer Revenue (OBSERVED)**: There is no active automated customer payment processor or verified fiat revenue ledger in the inspected codebase. In src/components/DonateModal.tsx, credit/debit card processing via Stripe is hardcoded as inactive (const stripeComingSoon = true; with a "Coming Soon" badge), while donation buttons link externally to off-platform PayPal (https://paypal.me/jstrongman47) and Cash App (https://cash.app/$Rainbowjobs94) accounts. In src/components/platform/CalendarPage.tsx, event ticket prices are populated from static mock data (mockEvents in @/lib/mockData) without payment settlement hooks.
- **Simulated Token Valuation & Assumed Prices (OBSERVED)**: In src/components/platform/FileMiningTab.tsx, the Shift Coin Protocol to Reality Coin (RC) valuation calculator is explicitly annotated in the code as a simulation: // Simulated blockchain data - in production this would come from actual protocol. It calculates a nominal value from user-supplied file size (S_{\text{MB}}) using static constants (COST_PER_MB = 0.50, currentBlockSpeed = 1247, shiftCoinRate = 0.0847, nearestNodeFactor = 1.15): \text{BaseCost} = 0.50 \times S_{\text{MB}}, \quad \text{SmartContractValue} = \frac{S_{\text{MB}} \times 0.0847 \times \left(\frac{\log_{10}(1247 + 1)}{3}\right)}{1.15} The UI labels the sum (totalRealityCoin) as Mining to Reality Coin (RC) at $1:1 USD and displays a 75% creator share (creatorEarnings) and 25% platform fee (platformFee). In the verified codebase, these figures are front-end UI calculations based on an assumed 1:1 USD peg and do not represent collected fiat revenue, backed stablecoin reserves, or verified on-chain token issuance.
- **Internal Database Balances (OBSERVED)**: In supabase/migrations/20260223161346_f62f7ea4-fd35-478c-b299-12ef31abd5d0.sql and src/hooks/usePostMining.ts, the public.post_mining_content table stores price_per_view (default 1.00), total_views (default 0), and total_earnings (default 0) as standard PostgreSQL numeric columns, while src/components/platform/RevenueModelSelector.tsx allows creators to select nominal split rules (75% Creator / 25% Platform for creator-first;

50% Viewers / 25% Creator / 25% Platform for viewer-rewards). These database records track internal application state rather than escrowed deposits or settled payouts.

2. Commercial & Economic Design Suggestions ([PROPOSAL — NOT APPROVED ECONOMICS])

- **[PROPOSAL — NOT APPROVED ECONOMICS] Decouple File Byte Size from Unbacked USD/Token Issuance**: Rewarding uploads proportionally to raw file size ($0.50 × MB) without collected customer payment creates an immediate Sybil and storage-bloat vulnerability where users upload maximum-sized (500 MB) dummy files to claim nominal credits while increasing cloud storage costs (postmining bucket). Any reward distribution should be bounded by verified, realized net customer revenue rather than unbacked per-megabyte issuance.
- **[PROPOSAL — NOT APPROVED ECONOMICS] Separate Accounting Ledgers**: Maintain strict ledger separation in PostgreSQL between: (a) **Settled Fiat Customer Revenue** (verified via webhook receipts from a live payment processor after fees/chargebacks), (b) **Escrowed User Deposits / Prepayments**, (c) **Non-Redeemable Internal Engagement Points**, and (d) **On-Chain Token Emissions** (governed by a fixed, capped emission schedule with no promised USD floor price).
- **[PROPOSAL — NOT APPROVED ECONOMICS] Cryptographic Nullifiers Before Reward Template Reuse**: Even after genuine compute and storage proofs are integrated, every verified proof submission must record a unique, canonical nullifier (hash(circuit_id, challenge_epoch, work_commitment, canonical_public_inputs)) in a uniqueness constraint table so that neither identical proofs nor malleable encodings of the same work can trigger duplicate reward accounting.

Unresolved Access Limitations and Search Gaps

1. **Missing docs/mining/* and scripts/report-template-proof-lab/* Tree**: The five target files referenced in your prompt are not included in ReplitExport.zip, ReplitExport-2.zip, or rainbowjobs94-ltbeta-e6afdc26-8a5edab282632443_1772017269522.txt, nor are they mounted in the local execution container.
2. **Truncated Export Chunks in Drive**: While the indexed portions of ReplitExport.zip (21 parts), ReplitExport-2.zip (22 parts), and rainbowjobs94-ltbeta-e6afdc26...txt (96 parts) expose the root package.json, vite.config.ts, Supabase SQL migrations, and React platform components (FileMiningTab.tsx, usePostMining.ts, RevenueModelSelector.tsx, DonateModal.tsx), none of those archives contain a scripts/report-template-proof-lab/ or docs/mining/ directory.
3. **How to Resolve**: Upload the specific archive or files (docs/mining/report-template-proof-validation.md, scripts/report-template-proof-lab/policy.test.mjs, docs/mining/candidate-proof-system-design.md, docs/mining/proof-artifact-cloud-search.md, and docs/mining/proof-and-sustainable-economics-research-brief.md) to your authorized Google Drive workspace or paste their contents directly into our conversation so they can be inspected and verified.

Would you like me to do a more in-depth research instead and propose a research plan?